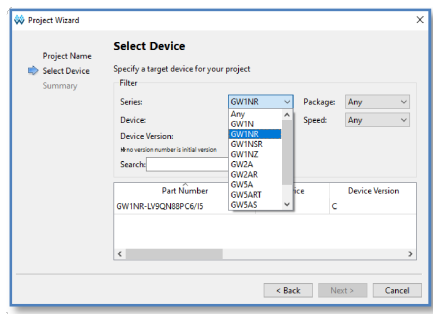
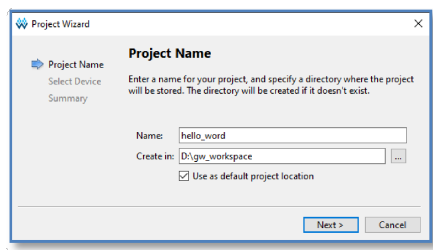
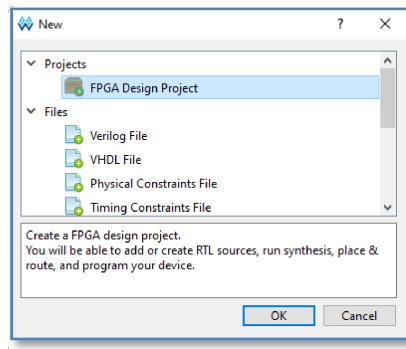
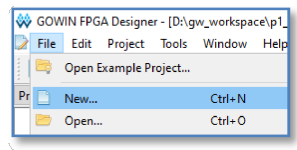


Setup Gowin for TangNano9K



https://gowinsemi.com/en/support/download_eda/



Tang Nano 9K

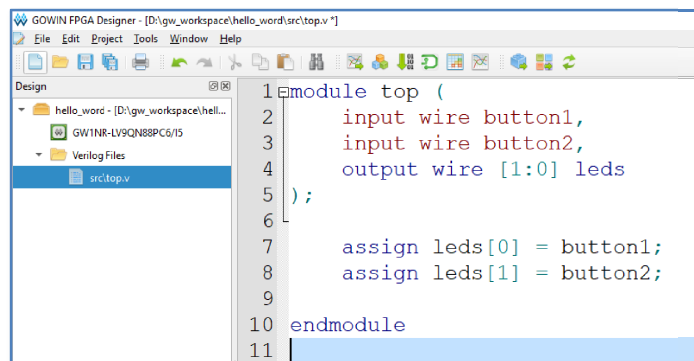
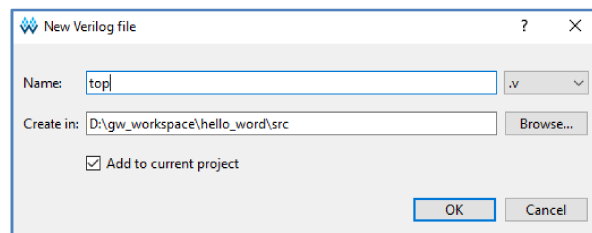
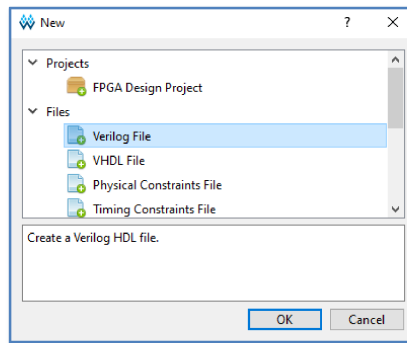
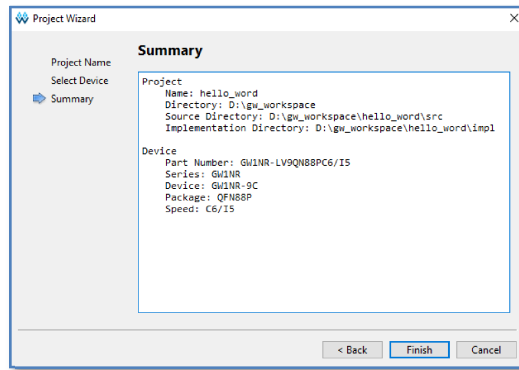
Series: **GW1NR**
Device: **GW1NR -9C**
Part Number: **GW1NR-LV9QN88PC6/I5**

Tang Nano 20K

Series: **GW2AR**
Device: **GW2AR-18**
Part Number: **GW2AR-LV18QN88C8/I7**

Tang Primer 20K

Series: **GW2A**
Device: **GW2A-18**
Part Number: **GW2A-LV18PG256C8/I7**



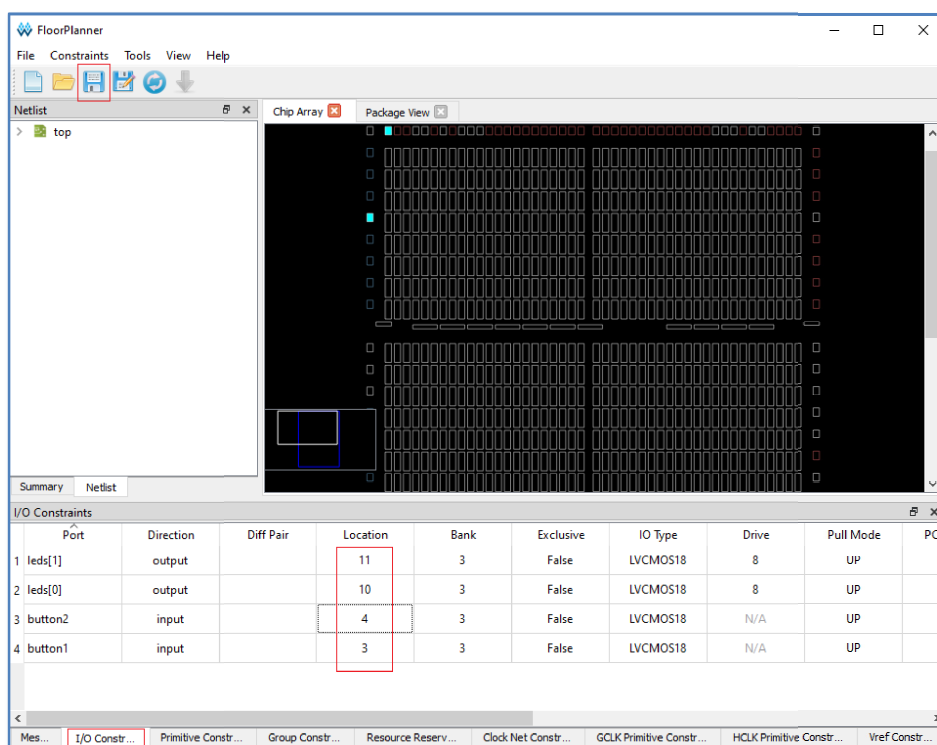
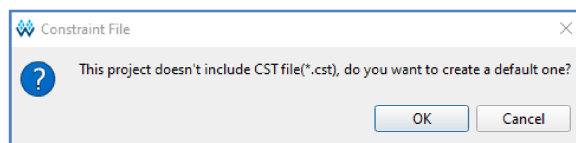
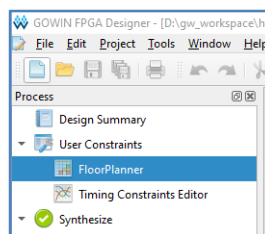
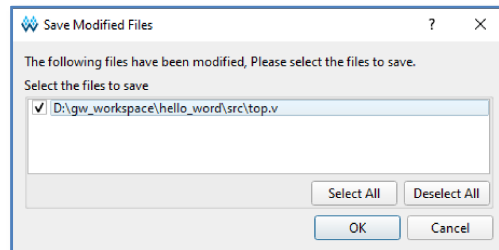
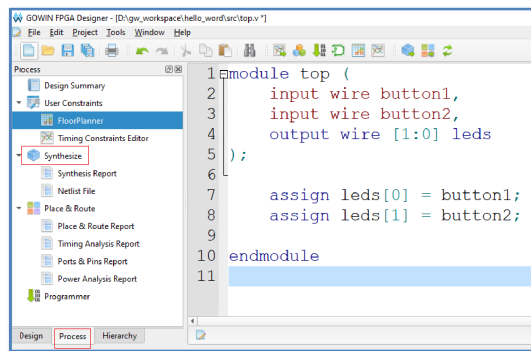
top.v

```

module top (
    input wire button1,
    input wire button2,
    output wire [1:0] leds
);

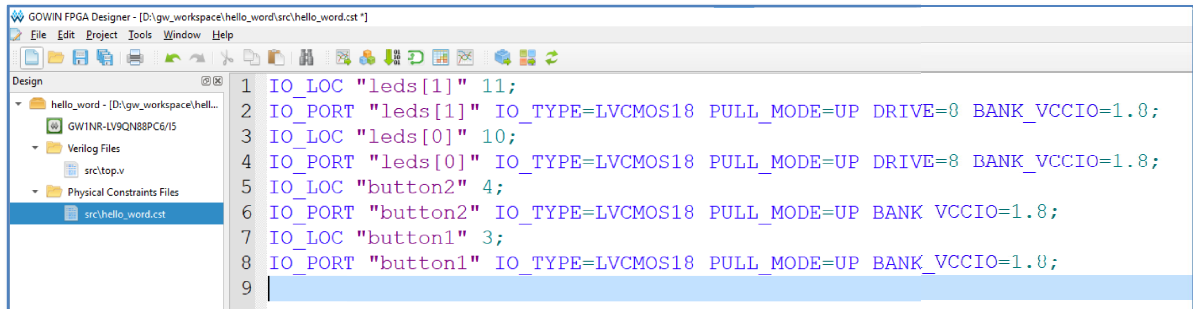
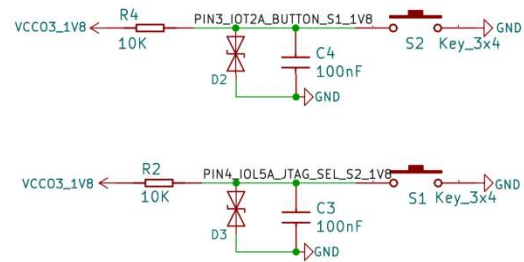
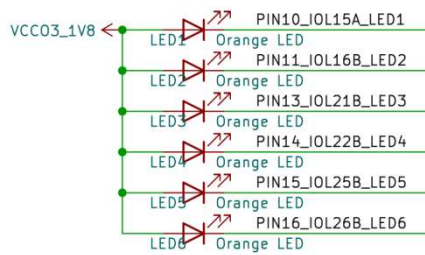
    assign leds[0] = button1;
    assign leds[1] = button2;

endmodule
  
```



Hardware Schematic

<https://wiki.sipeed.com/hardware/en/tang/Tang-Nano-9K/Nano-9K.html>

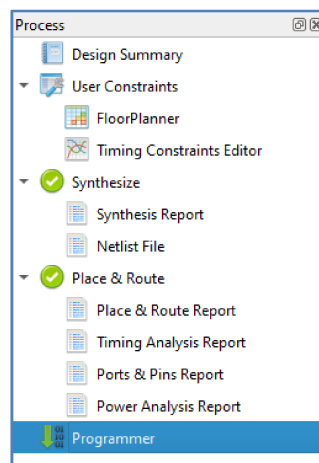
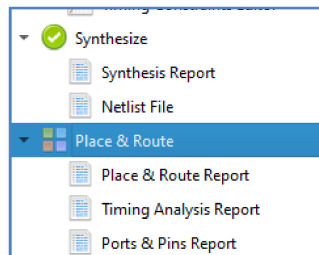


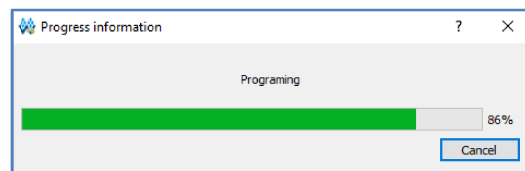
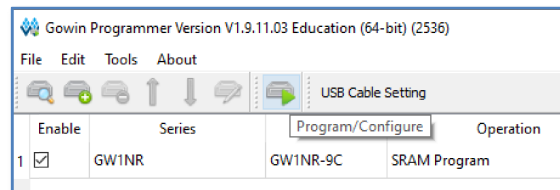
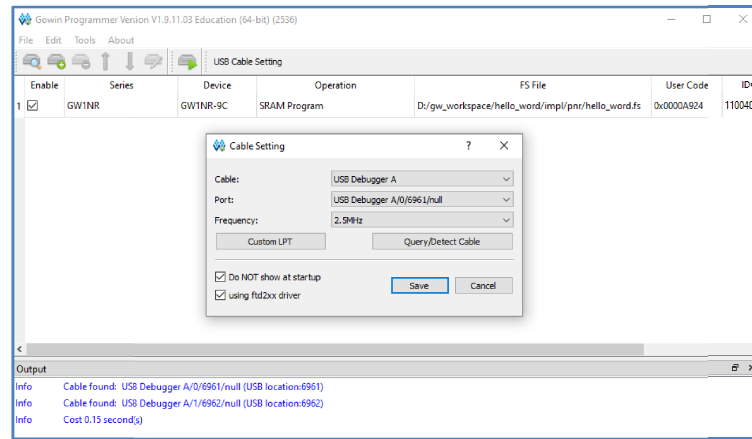
hello_word.cst

```

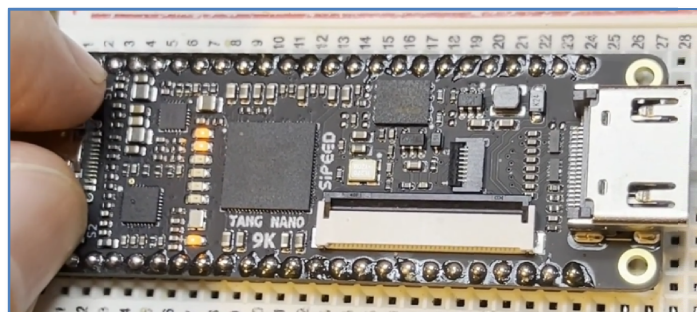
IO_LOC "leds[1]" 11;
IO_PORT "leds[1]" IO_TYPE=LVC MOS18 PULL_MODE=UP DRIVE=8 BANK_VCCIO=1.8;
IO_LOC "leds[0]" 10;
IO_PORT "leds[0]" IO_TYPE=LVC MOS18 PULL_MODE=UP DRIVE=8 BANK_VCCIO=1.8;
IO_LOC "button2" 4;
IO_PORT "button2" IO_TYPE=LVC MOS18 PULL_MODE=UP BANK_VCCIO=1.8;
IO_LOC "button1" 3;
IO_PORT "button1" IO_TYPE=LVC MOS18 PULL_MODE=UP BANK_VCCIO=1.8;

```



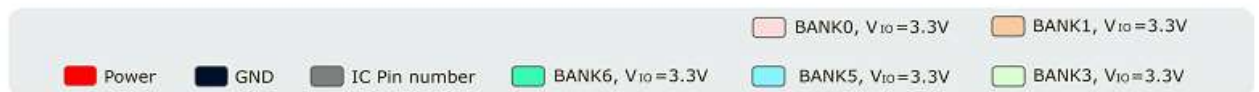
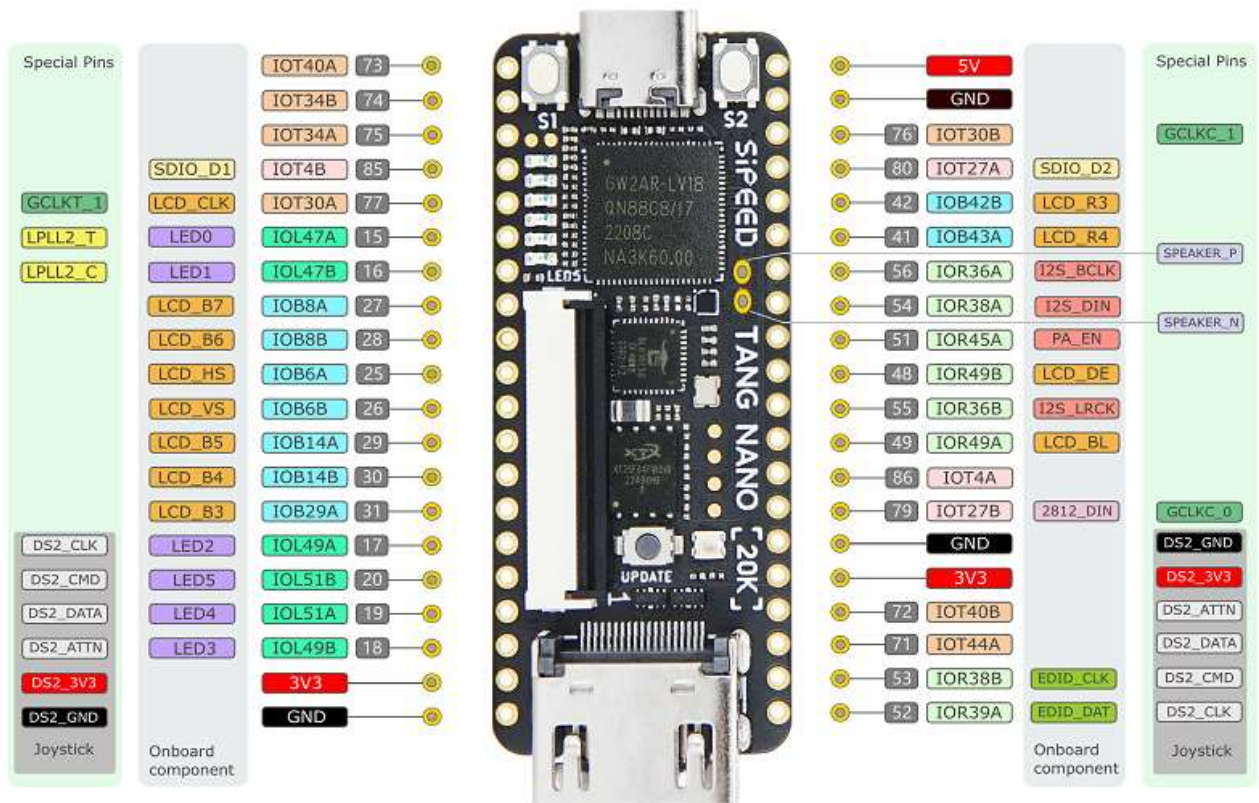


Tang Nano 9K
GW1NR-LV9QN88PC6/15



Tang Nano 20K
GW2AR-LV18QN88C8/I7





27 MHz: 4 (3.3V) not gclkt!
 Button1: 88 (3.3V)
 Button2: 87 (1 when pressed)
 LED[0]: 15 (3.3V)
 LED[1]: 16 (0 lights them)
 LED[2]: 17
 LED[3]: 18
 LED[4]: 19
 LED[5]: 20
 UART_TX: 69 (3.3V)
 UART_RX: 70
 WS2812 DIN: 79 (3.3V!)
 PLL_CLK0: 10 (3.3V) is GCLKT_6
 PLL_CLK1: 11
PLL_CLK2: 13 connects to BL616
 GCLKT_1: 77 (good for ext clock)

Y = good pin to use

<https://wiki.sipeed.com/hardware/en/tang/tang-nano-20k/nano-20k.html>

Яка різниця між Tang 9K та 20K

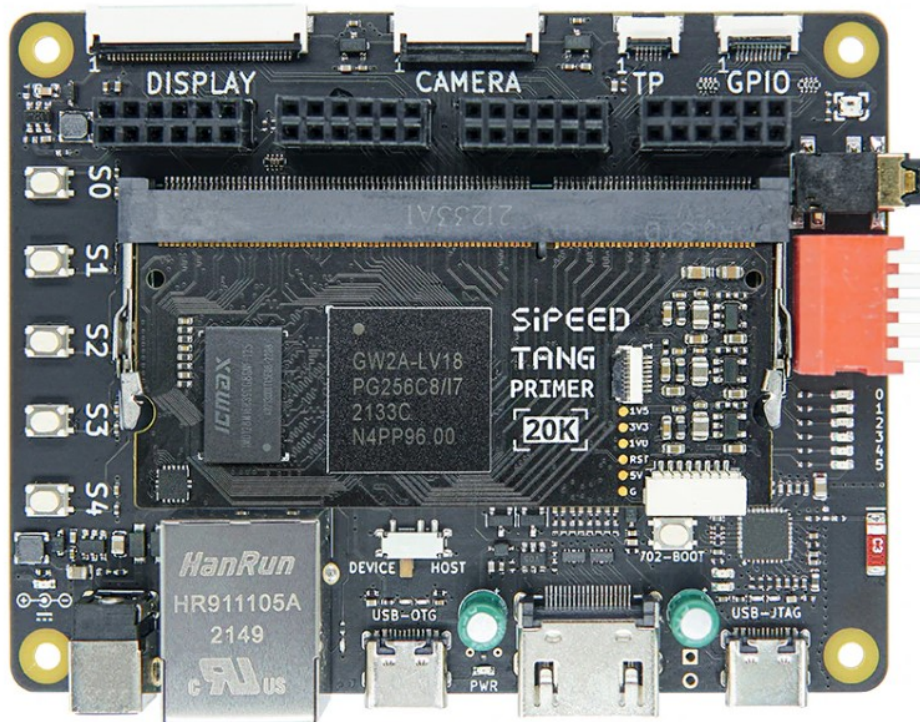
<https://hackaday.com/2024/06/06/whats-the-difference-between-tang-9k-and-20k-it-isnt-11/>

https://github.com/grughuhler/picorv32_tang_nano_20k

Tang Nano 20K vs Tang Nano 9K FPGA Board

<https://www.youtube.com/watch?v=I-N0brdlzzk>

Tang Primer 20K
GW2A-LV18PG256C8/I7



Burn without DOCK ext-board

Tang Primer 20K CM board can download bistream by RV debugger plus with connecting cable like below.

CM Board	5V0	TMS	TDO	TCK	TDI	RX	TX	GND
Debugger	5V0	TMS	TDO	TCK	TDI	TX	RX	GND



Обзор платы Tang Primer 20K на Gowin GW2A-LV18
<https://habr.com/ru/companies/timeweb/articles/747346/>